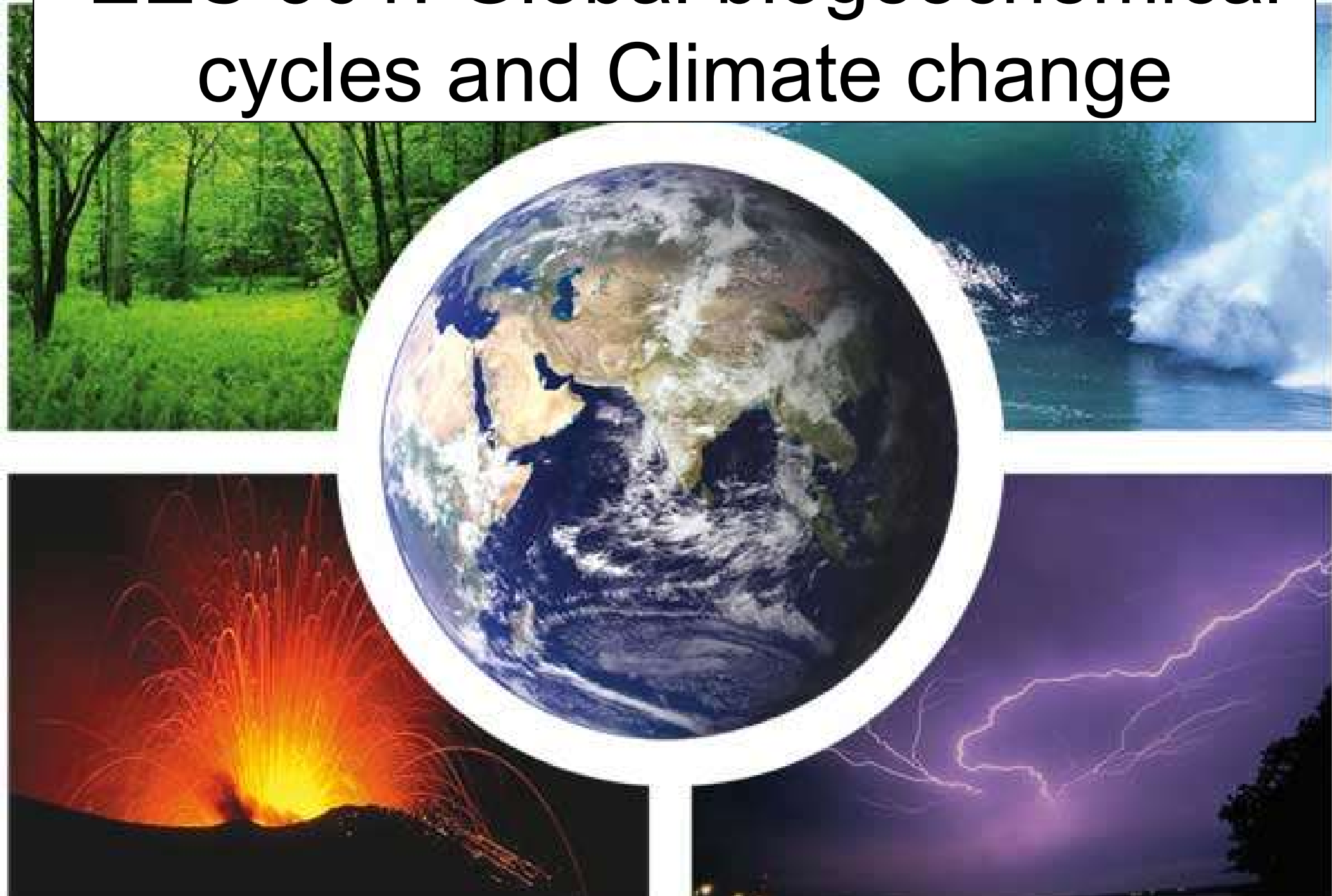


EES 301: Global biogeochemical cycles and Climate change



The climate system

- Climate is nothing but the long term average weather and defined by long term average temperature, pressure and precipitation on Earth.
- In a more local context we speak of “Climate zones” or the climate of a place. The climate of a place actually defines what can grow where depending on the parameters temperature, pressure, and rainfall in relation to potential evapotranspiration (the amount of water a plant will need to grow at that temperature while keeping stomata open for gas exchange)

Tropical moist climate



Arid climate



Temperate climate

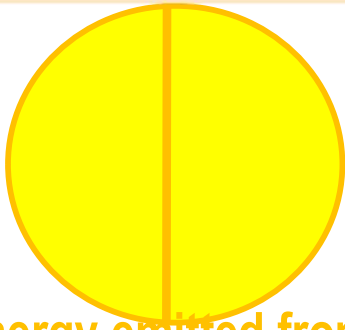


Radiative equilibrium for Earth

$$\frac{P}{A} = \sigma T^4 \text{ j/m}^2\text{s} \quad \text{Stefan-Boltzmann Law}$$

$$\sigma = 5.6703 \times 10^{-8} \text{ watt / m}^2\text{K}^4$$

This solar luminosity is then divided by the surface area of a sphere with the Sun Earth distance $4\pi d_{S-E}^2 = 4\pi(149.6 \times 10^9)^2 = 2.81 \times 10^{23} [\text{m}^2]$; $3.85 \times 10^{26} [\text{W}] / 2.81 \times 10^{23} [\text{m}^2] = 1370 [\text{W/m}^2] = S_0$



The energy emitted from sun is calculated based on Boltzman's law $5.6703 \times 10^{-8} \times 4 \times \pi \times 696,400,000^2 \times 5778^4 = 3.85 \times 10^{26} [\text{W}]$

Only half of the tilted sphere called earth is pointing towards sun at any point in time and the light intercepted is proportional to the area of a flat disk facing sun $= \pi R_e^2$ so $1370 [\text{W/m}^2] \times 6,371,000^2 [\text{m}^2] = 1.75 \times 10^{17} [\text{W}]$ of energy is intercepted

Energy absorbed = Energy emitted

$$\pi R_e^2 (1 - A) S_0 = \varepsilon 4\pi R_e^2 \sigma T_e^4$$

R_e = Radius of the Earth
 S_0 = Solar flux
 A = Albedo (~0.3 at present)

The albedo of a surface is a material property related to the color of the material.

Thanks to the albedo only $\pi R_e^2 (1 - A) S_0 = 240 \text{ W/m}^2$ radiation is absorbed by Earth.

Surface	albedo
Fresh snow	0.8-0.95
Old snow	0.42-0.70
Sandy soils	0.25-0.45
Clay soils	0.20-0.35
Peat soils	0.05-0.15

What would happen if melting of ice caps reduced albedo to 0.25?
 The amount of radiation absorbed would increase to 257 W/m^2 unless higher cloud cover due to an accelerated water cycle offsets some of the change

Radiative equilibrium for Earth

$$\frac{P}{A} = \sigma T^4 \text{ J/m}^2\text{s} \quad \text{Stefan-Boltzmann Law}$$

$$\sigma = 5.6703 \times 10^{-8} \text{ watt / m}^2\text{K}^4$$



R_e = Radius of the Earth
 S_0 = Solar flux
 A = Albedo (~ 0.3 at present)

Energy absorbed = Energy emitted

$$\pi R_e^2 (1 - A) S_0 = \varepsilon 4\pi R_e^2 \sigma T_e^4$$

240 W/m²

For the planet to be in equilibrium the amount of energy emitted must be equal to the amount of energy absorbed. Emissivity $\varepsilon = 1$ at the top of the atmosphere)

We can calculate at which black body temperature of Earth T_e this would be the case. For this we solve $(1-A)S_0 = 4\sigma T_e^4$

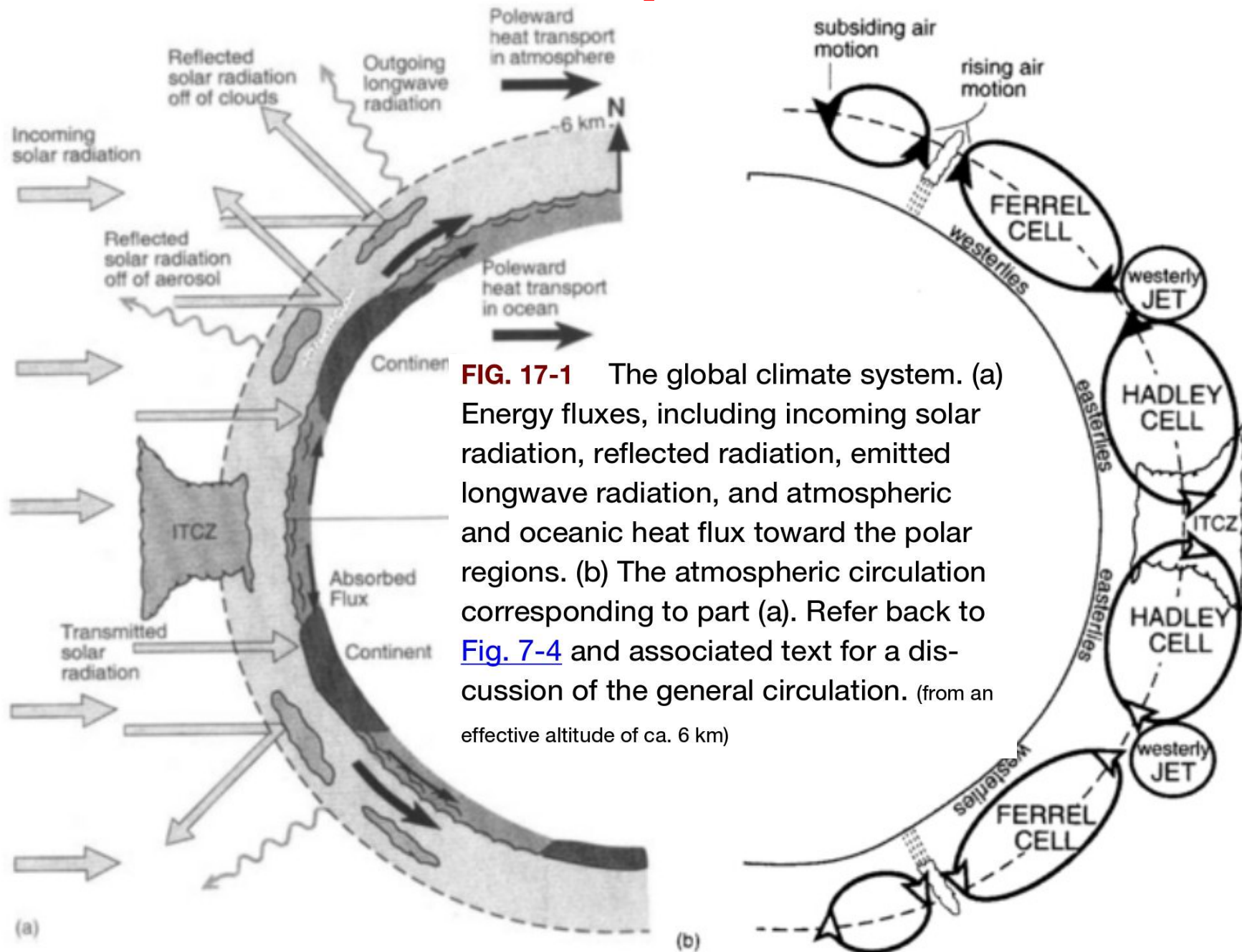
$$T_e = \left[\frac{S_0(1-A)}{4\sigma} \right]^{1/4}$$

$$\{1370 \text{ [W/m}^2\text{]}/4/(5.6703 \times 10^{-8} \text{ [K}^{-4}\text{W/m}^2\text{]})\}^{1/4}$$

255 K

The equilibrium temperature of Earth 255 K is well below the freezing point of water and the global mean surface temperature which is 288K. Instead this temperature is attained at an altitude of 6 km, from which the black body called Earth actually radiates most of its Energy to space. The surface temperature is controlled by greenhouse gas concentrations in the atmosphere.

Radiative equilibrium for Earth



Climate models: Atmosphere

An Adiabatic process is a type of thermodynamic process that occurs without transferring heat between the thermodynamic system and its environment. This includes all processes that transport energy from the surface to higher altitudes and from the equator to the pole.

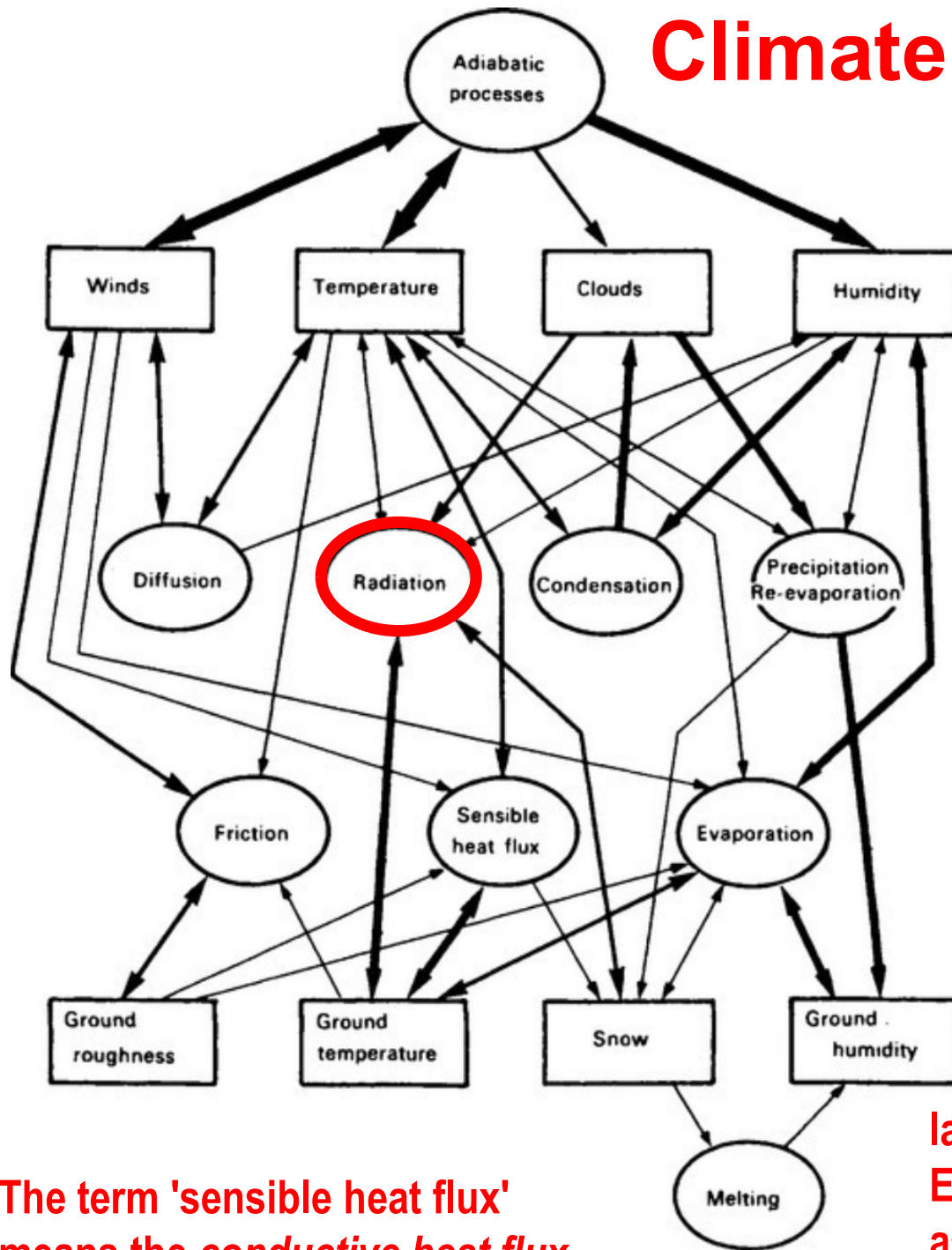
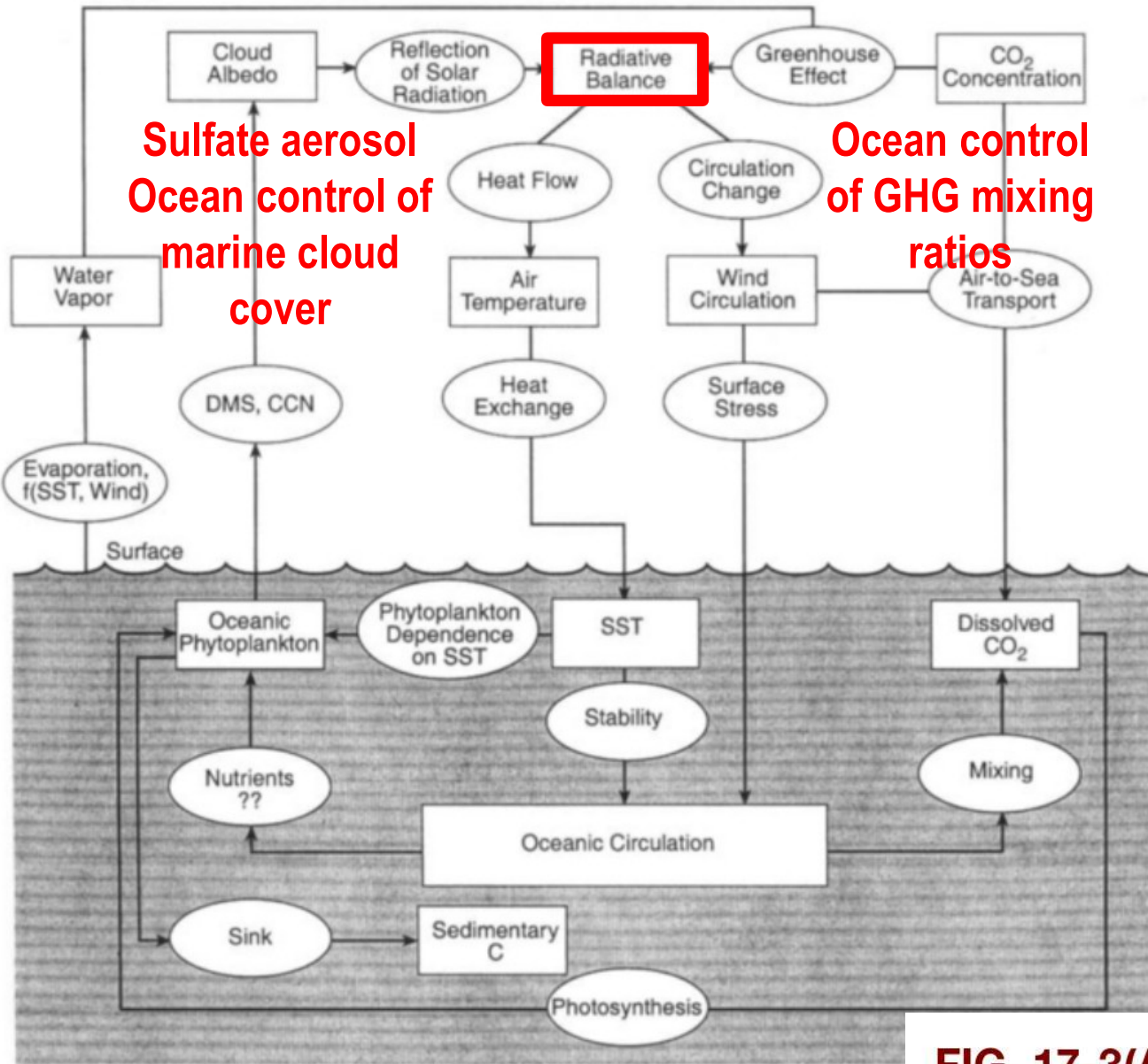


FIG. 17-2 The web of interactions in the atmospheric part of the global climate system. The strength of the interactions is qualitatively depicted by the thickness of the line. Bidirectional interactions have two arrowheads, unidirectional ones have only one. (From [Houghton \(1984\)](#), reprinted with permission from Cambridge University Press.)

The term 'sensible heat flux' means the *conductive heat flux from the Earth's surface to the atmosphere where it changes T*

latent heat flux is the flux of energy from the Earth's surface to the atmosphere that is associated with evaporation or transpiration of water at the surface and subsequent condensation of water vapor in the troposphere

Climate models: Ocean

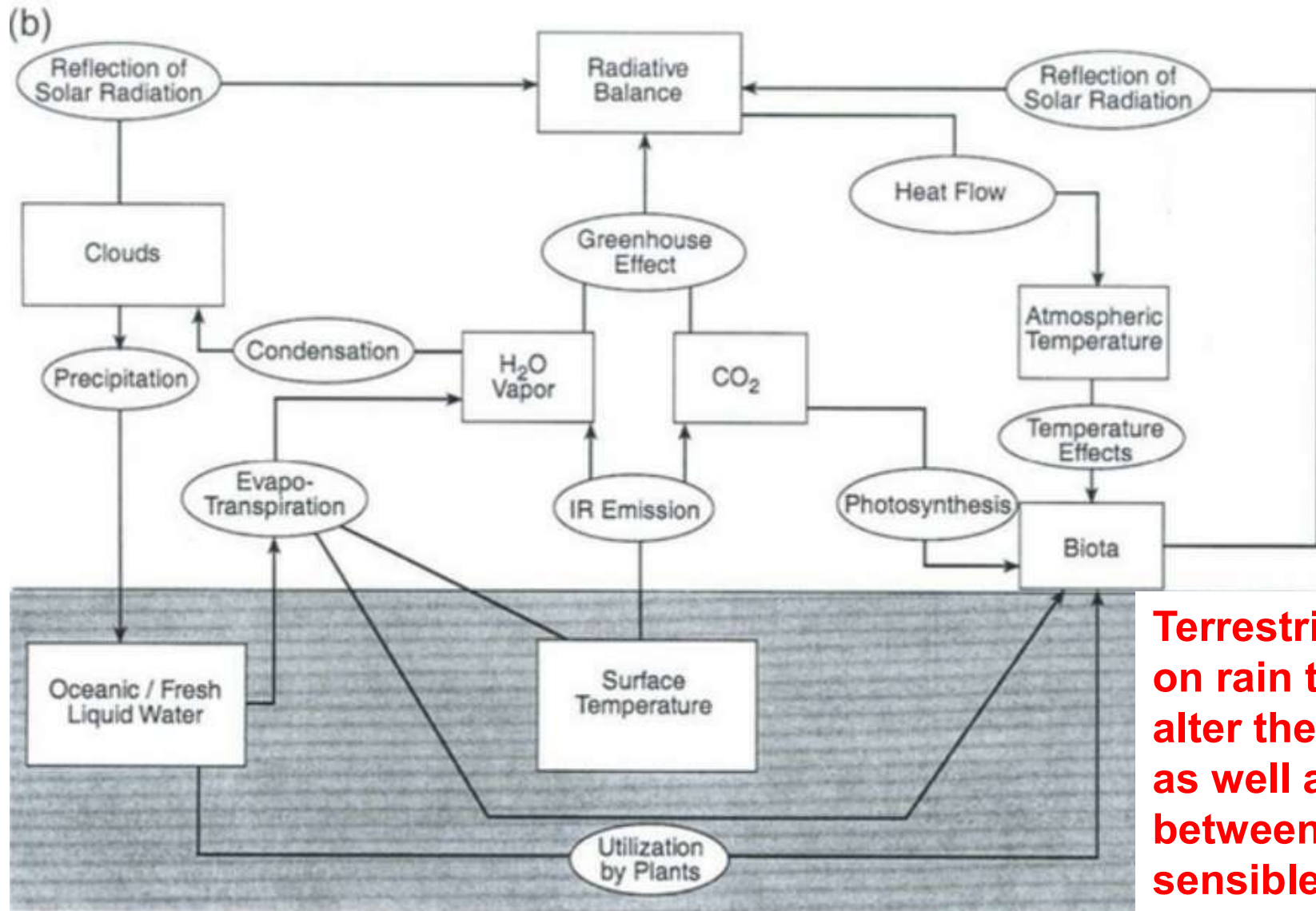


1. The ocean plays a key role in controlling both the concentration of the natural GHG water vapor and in sequestering some of the anthropogenic CO₂

2. Phytoplankton controls most of the natural sulfur cycle and with it natural cloud cover as the sulfate aerosol formed from DMS oxidation products provides the most important CCN over the ocean.

FIG. 17-3(A) A simplified picture of the coupling of biogeochemistry and global radiative balance in marine systems. SST = Sea surface temperature.

Climate models: Terrestrial biosphere



Terrestrial biota depend on rain to survive. They alter the surface albedo as well as the split between latent and sensible heat fluxes to the atmosphere. They also contribute to sequestering CO₂ and emit CCN and IN

FIG. 17-3(B) Coupling of biogeochemistry and global radiative balance in terrestrial systems.

The dynamics of the climate system: Forcings

Forcings given in W/m^2 are externally imposed changes in the energy balance of the planet. Forcings play the role of the initial stimulus that perturbs the variable temperature and triggers several regulatory loops within the climate system.

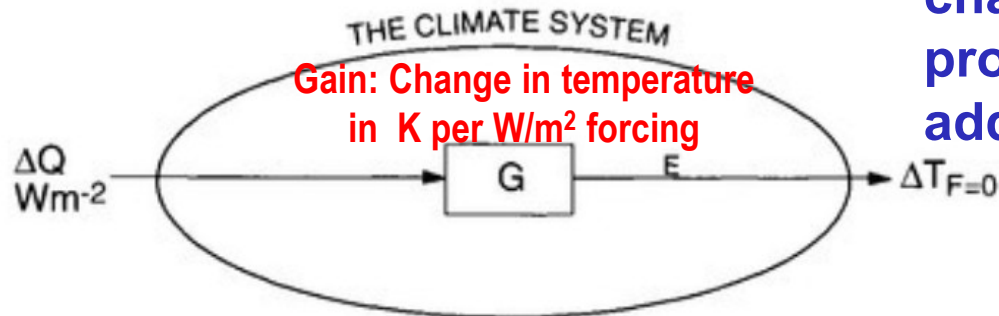
Astronomical changes like the warming of the sun (30% over 4.5 Ga), and changes in the orbital parameters of the earth (which happen on a $10^4 - 10^5$ timescale) can alter the climate. However, these act on a very long time scale (e.g. $0.09 \text{ W m}^{-2} \text{ Myr}^{-1}$ for warming of the sun) compared to the current anthropogenic forcing ($2\text{-}3 \text{ W m}^{-2} \text{ century}^{-1}$). Orbital forcings are known as Milankovich effect.

Major volcanic eruptions or meteorite impacts can impose forcings to the climate by putting a large amount of dust into the atmosphere for a couple of years to decades. The mass extinction of the dinosaurs is believed to have been triggered by such a asteroid impact related dust injection that suppressed photosynthetic activity.

Changes in the plate tectonic system such as the eruption of trapp basalts or changes in the sea floor spreading rate can impact the climate system by raising atmospheric CO_2 levels to $>2000 \text{ ppm}$. The Permian-Triassic mass extinction event during which the eruption of Siberian Trapp basalts caused euxinia (oxygen-starved, sulfurous and acidic oceans) is believed to have been triggered such a geological factor.

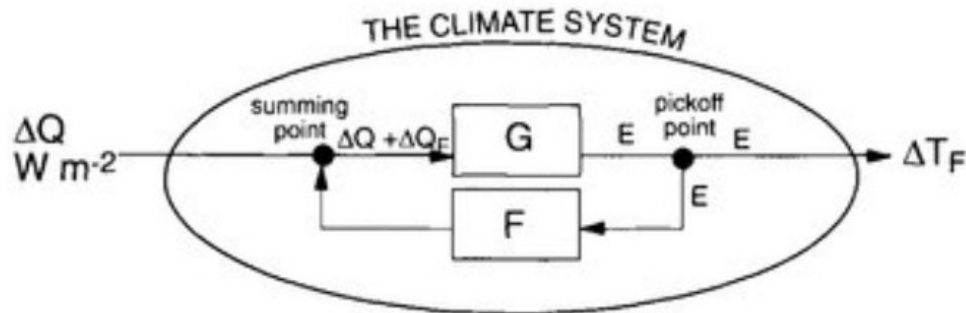
The dynamics of the climate system: Feedbacks

(a) No feedback



In a climate system with no feedbacks the change in temperature would be directly proportional to the change in forcing with no additional forcing

(b) With feedbacks that are internal to the climate system



In a climate system with feedbacks a change in temperature can trigger an internal feedback loop which results in an additional greenhouse effect caused e.g. by melting ice caps and changes in albedo. The total forcing is now the original forcing plus the additional effective forcing due to the feedback

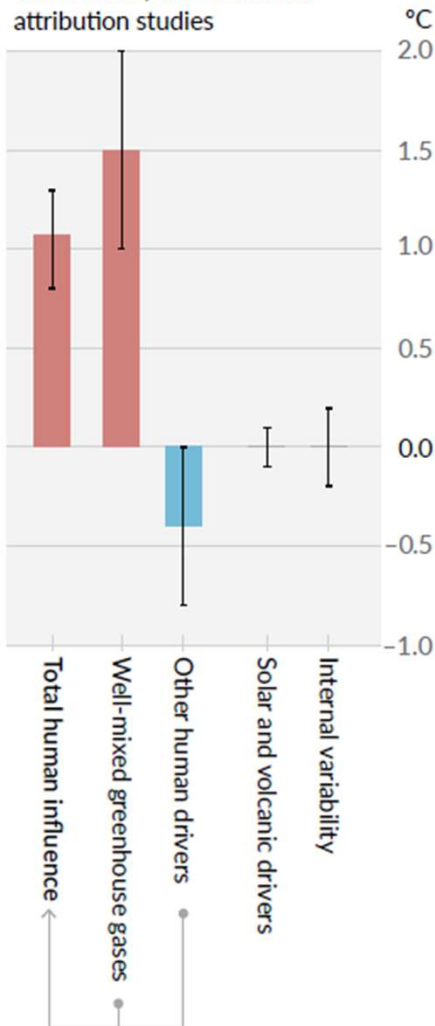
Where: E = the physical result of the gain operating on $\Delta Q + \Delta Q_F$
 ΔQ = forcing, $W m^{-2}$
 ΔQ_F = effective additional forcing due to feedback (can be positive or negative)
 G = gain. In this simple climate system G has units $K (W m^{-2})^{-1}$
 F = feedback ($W m^{-2} K^{-1}$)
 ΔT_F = temperature response with feedback, K .
 $\Delta T_{F=0}$ = temperature response with no feedback, K .

FIG. 17-4 Schematic of the climate system with and without feedbacks, that depend on temperature.

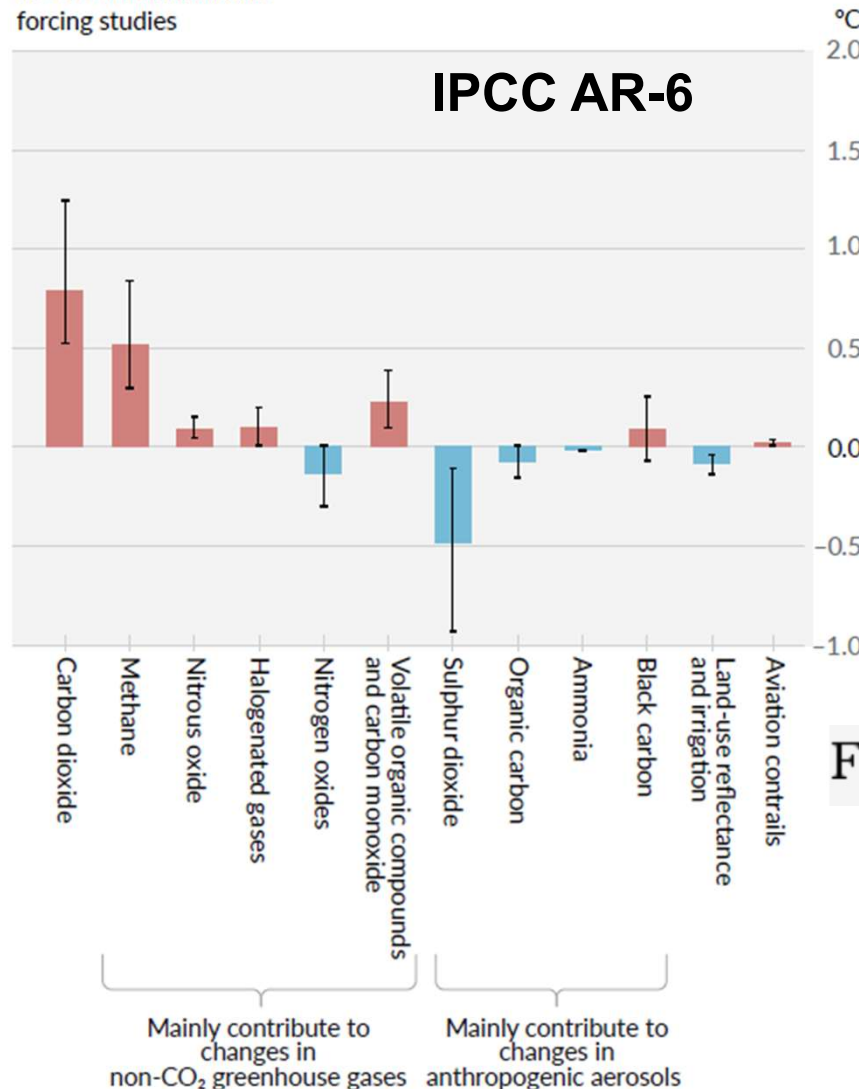
The complexity of quantifying forcings in the climate system

Contributions to warming based on two complementary approaches

(b) Aggregated contributions to 2010–2019 warming relative to 1850–1900, assessed from attribution studies



(c) Contributions to 2010–2019 warming relative to 1850–1900, assessed from radiative forcing studies



In a climate model forcings are defined as partial derivatives i.e. the forcing exerted on the system by one parameter with all other parameters being held constant at the value of the preindustrial reference period.

e.g. by putting only the anthropogenic CO₂ emissions into the model and

observing the resulting ΔT

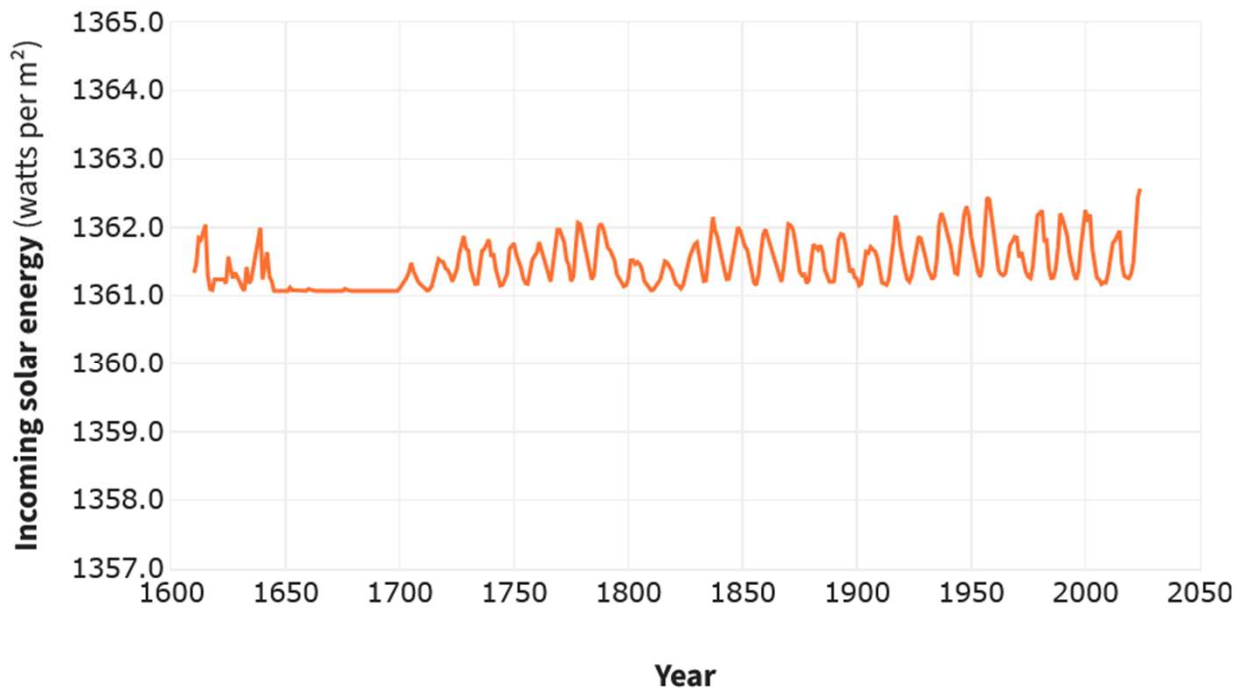
$$\text{Forcing} \equiv \int_{x_0}^{x_1} \frac{\partial Q}{\partial x} dx = Q(x_1) - Q(x_0)$$

These are generally reported in terms of the total change of the response variable temperature with respect to a reference period.

Natural climate forcings

The forcing from orbital changes (Milankovich cycles) is currently -0.035 W/m^2 per decade. Under ordinary circumstances, the next thing on our geological calendar would have been the next Ice age.

The solar output varies due to processes related to the interior of the sun and the change in output since the reference period is estimated at $0.3 \pm 0.2 \text{ W/m}^2$. The high uncertainty is related to the fact that both the reference period 1850-1900 and evaluation period 2010-2019 contain multiple stages of the sunspot cycle. In 2025 we are at the solar Maximum of cycle 25.



Large volcanic eruptions that put a lot of dust and SO_2 into the stratosphere from which aerosol is removed only slowly can decrease surface temperature by $\sim 0.5 \text{ K}$ for 1-3 years. The Mt. Pinatubo eruption had a peak forcing of -3 W/m^2 and average forcing of -1.5 W/m^2 over the period mid 1991-93.

Anthropogenic climate forcings

Trace gas	Radiative forcing approximation, ΔF in W/m^2
CO_2	$\Delta F = 6.3 \ln \left(\frac{[\text{CO}_2]}{[\text{CO}_2]_0} \right)$ where $[\text{CO}_2]$ is concentration in ppmv Valid for concentrations < 1000 ppmv
CH_4	$\Delta F = 0.036 \{ [M]^{1/2} - [M]_0^{1/2} \} - \{ f([M], [N]) - f([M]_0, [N]_0) \}$ where $[M]$ is concentration of CH_4 in ppbv and $[N]$ is concentration of N_2O in ppbv and $f([M], [N])$ is the methane–nitrous oxide overlap term ^a valid for $[M] < 5$ ppmv
N_2O	$\Delta F = 0.14 \{ [N]^{1/2} - [N]_0^{1/2} \} - \{ f([M]_0, [N]) - f([M]_0, [N]_0) \}$ with $[M]$ and $[N]$ as above valid for $[N] < 5$ ppmv
CFC-11	$\Delta F = 0.22 ([X] - [X]_0)$ where $[X]$ is concentration of CFC-11 in ppbv valid for $[X] < 2$ ppbv
CFC-12	$\Delta F = 0.28 ([Y] - [Y]_0)$ where $[Y]$ is concentration of CFC-12 in ppbv valid for $[Y] < 2$ ppbv
Stratospheric H_2O vapor From methane oxidation	$\Delta F = 0.011 ([M]^{1/2} - [M]_0^{1/2})$ with $[M]$ as above
Tropospheric O_3	$\Delta F = 0.02 ([O] - [O]_0)$ where $[O]$ is concentration of ozone in ppbv

Table 17-2

Current and projected global-mean anthropogenic greenhouse forcing, W/m^2

Greenhouse gas	1995	2025
CO_2	1.5	2.9
CH_4	0.4	0.7
N_2O	0.1	0.2
Chlorofluorocarbons	0.3	0.5
Total	2.3	4.6

Table 17-1

Expressions used to derive radiative forcing for past trends and future scenarios of greenhouse concentrations ([IPCC, 1990](#))

$$f([M], [N]) = 0.47 \ln [1 + 2.01 \times 10^{-5} ([M][N]^{0.75} + 5.31 \times 10^{-15} [M] ([M][N])^{1.52}), \text{ with } [M] \text{ and } [N] \text{ in ppbv.}$$

Anthropogenic climate forcings

➤ Different greenhouse gases absorb different wavelength ranges of the terrestrial radiation. Depending on how much or how little of the terrestrial radiation is at the right wavelength and how much of that wavelength is already absorbed by water vapour, their power to change global temperatures varies.

➤ Different gases have different lifetime. Some stick around relatively shortly e.g. methane (12.4 years) others for longer time periods e.g. CO₂ ~95 years, SF₆ 3200 years.

Global warming potential (GWP) is the heat absorbed by any greenhouse gas in the atmosphere, as a multiple of the heat that would be absorbed by the same mass of carbon dioxide (CO₂) being emitted at the same point in time. GWP is 1 for CO₂. For other gases it depends on the gas and the time frame of consideration. Some gases, have large GWP, since a ton of the gas absorbs much more heat than a ton of CO₂. Some gases, break down much faster than CO₂. They have a shorter atmospheric lifetime which can result in a large difference between their 20 year and 100 year GWP.

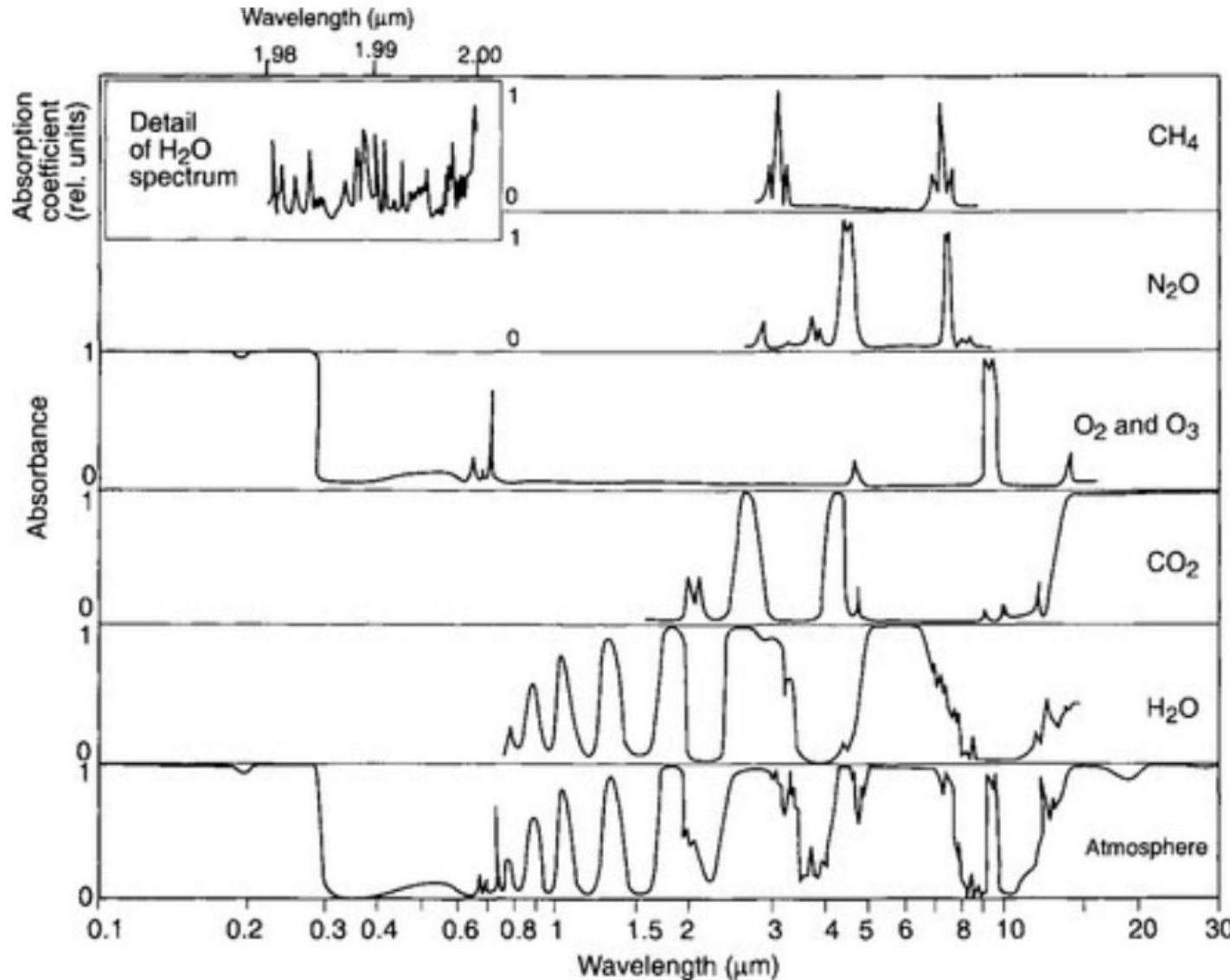
As a consequence of the short lifetime the heat absorption, or GWP, of methane over the next 20 years is a bigger multiple of CO₂ (96) than its heat absorption over 100 years (32). So in short when we calculate the GWP we look at 1 ton of CH₄ emitted now and ask how many tonnes of CO₂ would have to be emitted to do the same damage to the climate system as 1 ton of CH₄ does in 20 years (Answer: 96 tonnes of CO₂) or 100 years (Answer: 32 tonnes of CO₂) .

Anthropogenic climate forcings

GWP values and lifetimes ♦	Lifetime in years ♦	Global Warming Potential (GWP)			Source ♦
		20 years ♦	100 years ♦	500 years ♦	
Perfluorotributylamine (PFTBA)			7100		2013 GRL ^[3]
Methane		96	32		2018 Sci+2016 GRL ^{[4][5]}
Methane (biogenic)			39		2016 GRL with feedback ^[6]
Methane ("fossil")			40		2016 GRL with feedback ^[6]
Nitrous oxide	114	289	298	153	2007 p212 ^[2]
HFC-134a (hydrofluorocarbon)	14	3830	1430	435	2007 p212 ^[2]
CFC-11 (chlorofluorocarbon)	45.0	6730	4750	1620	2007 p212 ^[2]
Carbon tetrafluoride (CF ₄)	50000	5210	7390	1120	2007 p212 ^[2]
HFC-23 (hydrofluorocarbon)	270	12,000	14,800	12,200	2007 p212 ^[2]
Sulfur hexafluoride	3200	16,300	22,800	32,600	2007 p212 ^[2]

Compilation of GWP on Wikipedia. Some gasses we use have a much higher GWP than CO₂. This also means that some mitigation measures have hard to quantify CO₂ savings. E.g. Solar power may displace some coal or gas directly (direct & quantifiable CO₂ savings, ~ 3 kg per kg of avoided fuel combustion), but it also reduces emissions during the mining activity (e.g. coal bed methane leakage & CO₂ emissions from mining transport that are hard to quantify). In addition rooftop solar lowers the demand for power transmission capacity and hence SF₆ use (impossible to quantify because there is no separate database for installed rooftop capacity).

Anthropogenic climate forcings



A greenhouse gas (sometimes abbreviated GHG) is a gas that absorbs and emits radiant energy within the thermal infrared range.

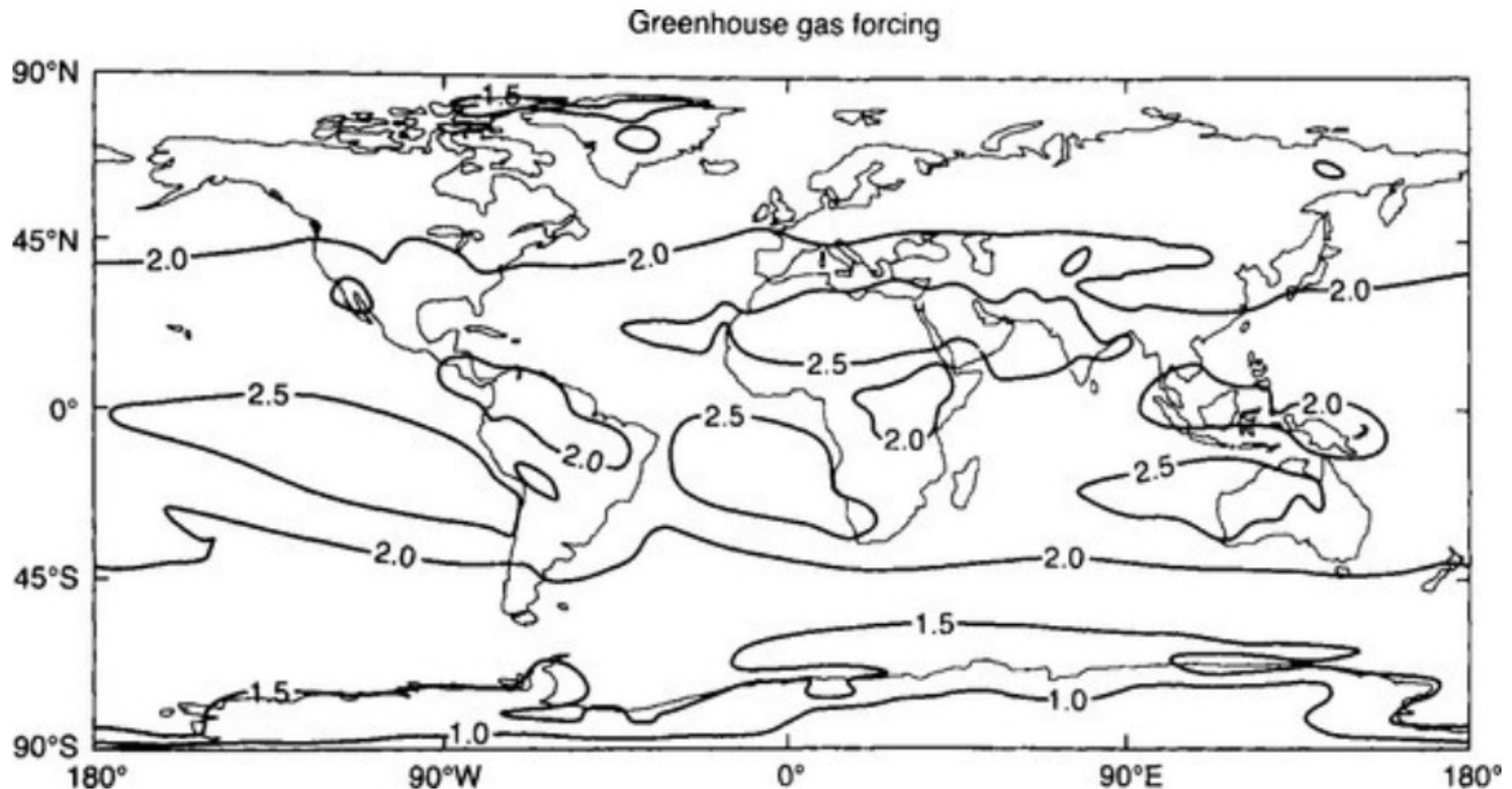
To act as anthropogenic greenhouse gas, a compound must absorb spectral lines that are not fully absorbed by water vapour already.

The most important anthropogenic greenhouse gasses are carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and tropospheric ozone (O_3).

The most important natural greenhouse gas in Earth's atmosphere is water vapor (H_2O).

The water vapour in the Earth's atmosphere is controlled by global sea surface temperature. Hence water vapour forcing is controlled by natural feedback cycles within in the earths climate system.

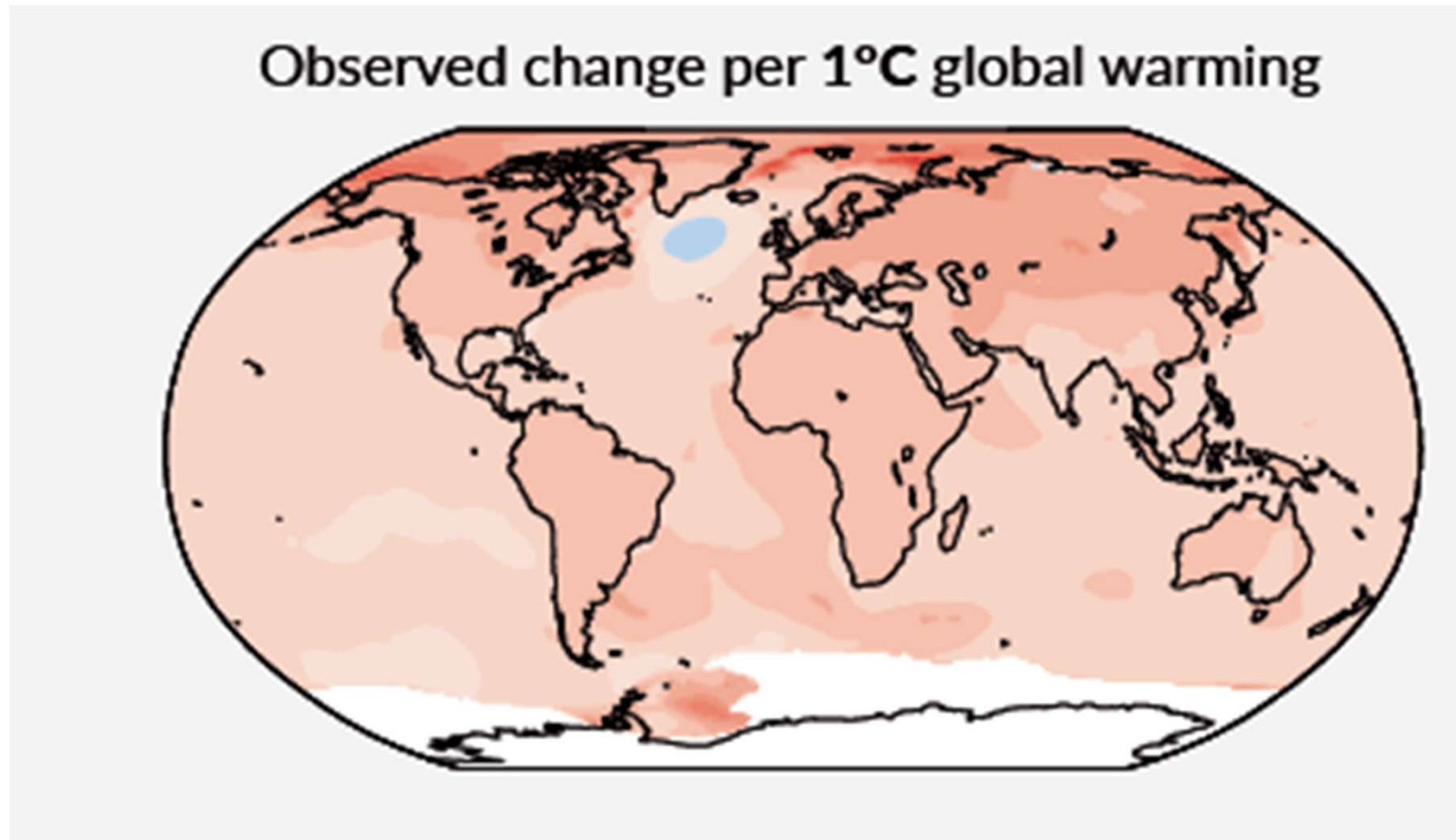
Anthropogenic climate forcings



Even though the greenhouse gasses are well mixed throughout the atmosphere the forcing exerted by them is not uniformly distributed. The forcing in W/m^2 is largest in the subtropics, where the air is dry due to spectral line overlap of GHG with water vapor. Forcings also depend on the upward flux of IR radiation which is proportional to T_e^4 and surface temperature is higher in the tropics than at the pole.

FIG. 17-6 Calculated geographical distribution of the climate forcing (W/m^2) by anthropogenic greenhouse gases alone, from pre-industrial periods to ca. 1990. See text for interpretation. From IPCC (1995), after [Kiehl and Briegleb \(1993\)](#). (Reprinted by permission from IPCC.)

Anthropogenic climate forcings



Even though the greenhouse gasses are well mixed throughout the atmosphere the forcing exerted by them is not uniformly distributed. The ΔT is largest in the Arctic ocean and near Antarctica where the sea ice melting feedback cycle amplifies the local temperature response. There is also a cooling south of Greenland where the NADP formation takes place, because the AMOC (Atlantic Meridional Overturning Circulation) is getting perturbed by melt water input and is slowing. This slows the energy transfer to the pole by the gulf stream.

Anthropogenic climate forcings

Aerosol particles influence the climate directly, by scattering incoming radiation. Atmospheric sulfate aerosol currently scatters 15-20% of the incoming solar radiation back to space. The sulfate scattering efficiency α at the global average humidity is $10 \text{ m}^2/\text{g}$. The rate at which light is lost from the incoming light beam is defined by the scattering coefficient σ . This quantity is defined by the aerosol mass M (g/m^3) multiplied with the scattering efficiency α

$$\sigma = M\alpha \text{ [m}^{-1}\text{]}$$

When both sides are integrated over the altitude z a dimensionless quantity called the aerosol optical depth δ results with respect to the average M in the vertical column in g/m^2 .

$$\int_0 \sigma dz = \delta = \alpha \int_0 M dz = \alpha B$$

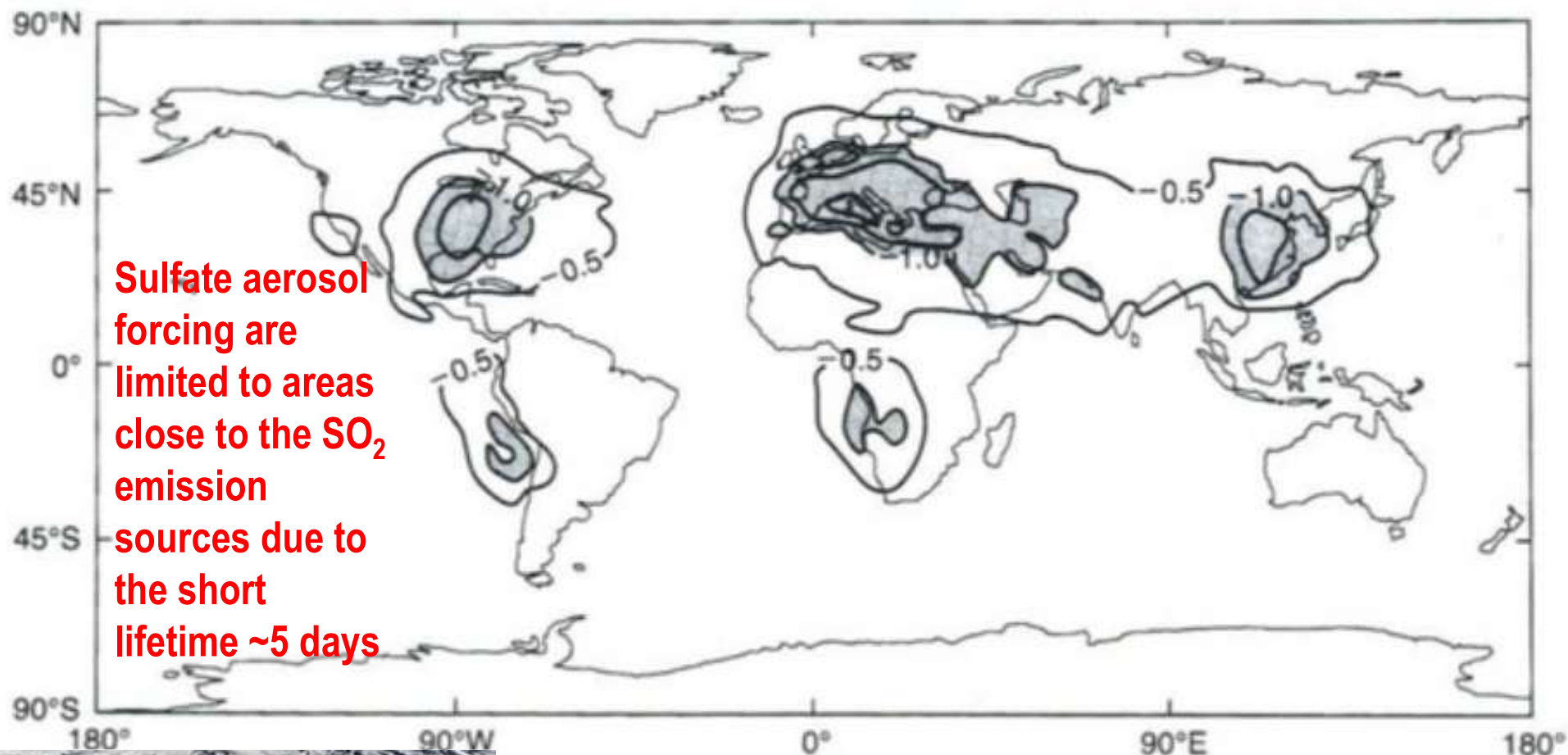
This is then used in Beer lamberts law where I is the transmitted radiation and I_0 the radiation received at the top of the atmosphere

$$\frac{I}{I_0} = e^{-\delta}$$

Aerosol particles also influence the climate indirectly, by changing cloud cover, cloud brightness and cloud lifetimes. Atmospheric sulfate aerosol currently scatters 15-20% of the incoming solar radiation back to space.

Anthropogenic climate forcings

Anthropogenic sulphate aerosol forcing



Sulfate aerosol forcing are limited to areas close to the SO₂ emission sources due to the short lifetime ~5 days



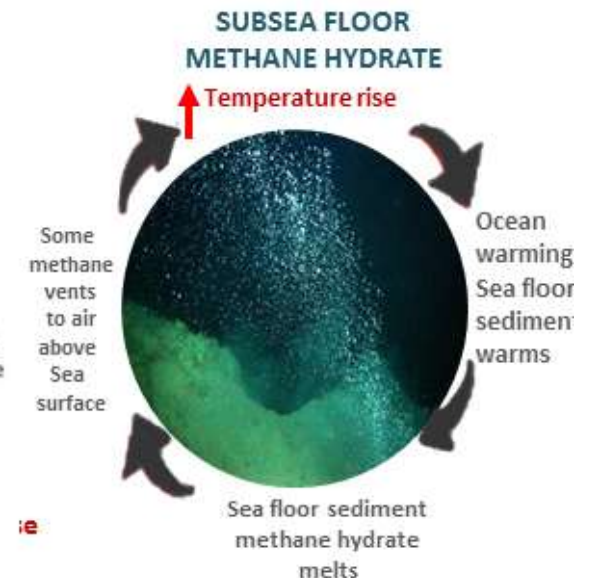
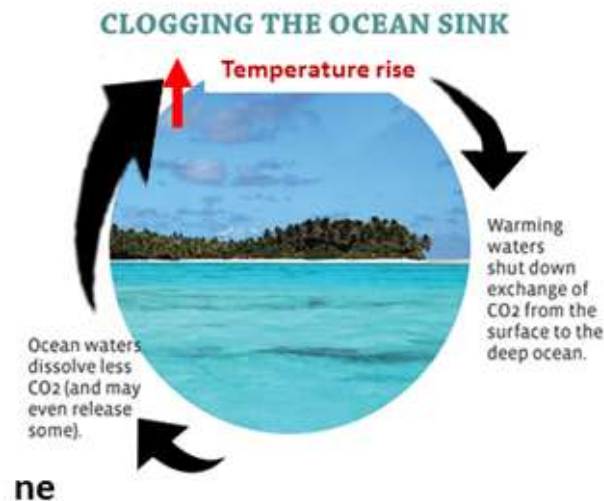
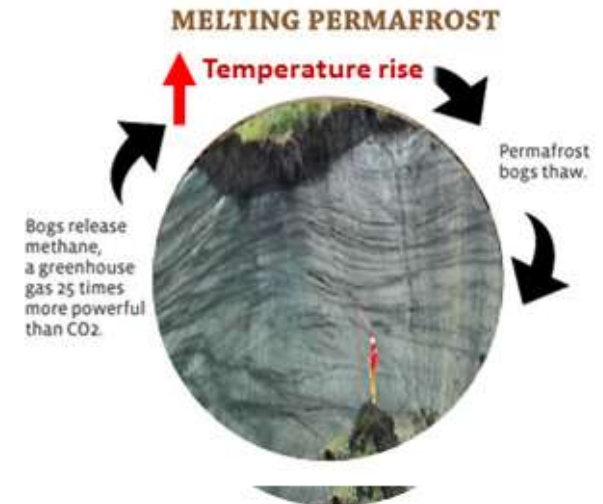
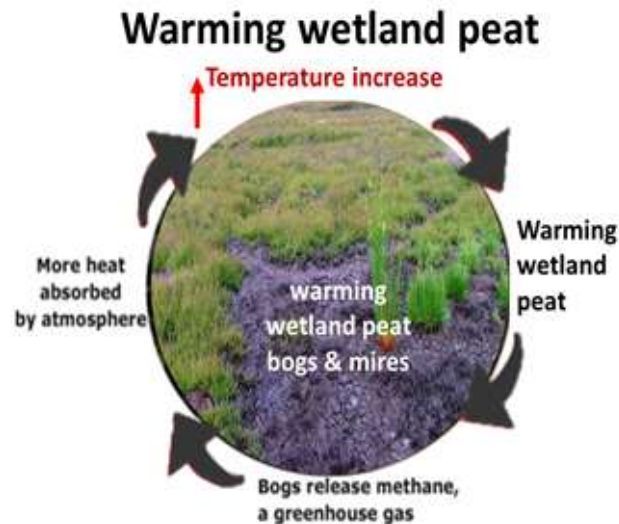
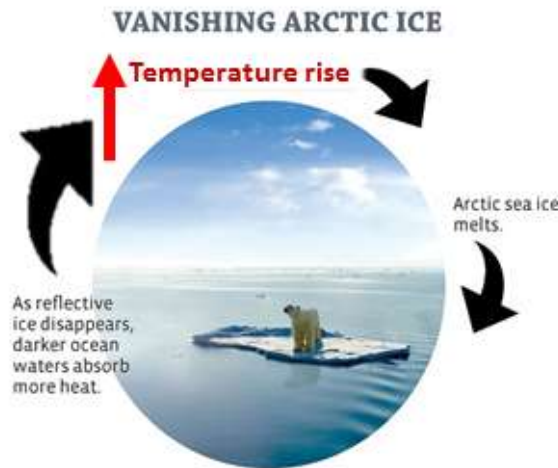
Ship tracks demonstrate the indirect effect on cloud brightness. The magnitude of the indirect aerosol effect is currently highly uncertain.

FIG. 17-7 Calculated geographical distribution of the direct climate forcing (W/m^2) by one anthropogenic aerosol component, sulfates, from pre-industrial periods to ca. 1990. (Reprinted with permission from [IPCC, 1995](#) .)

Feedbacks in the climate system

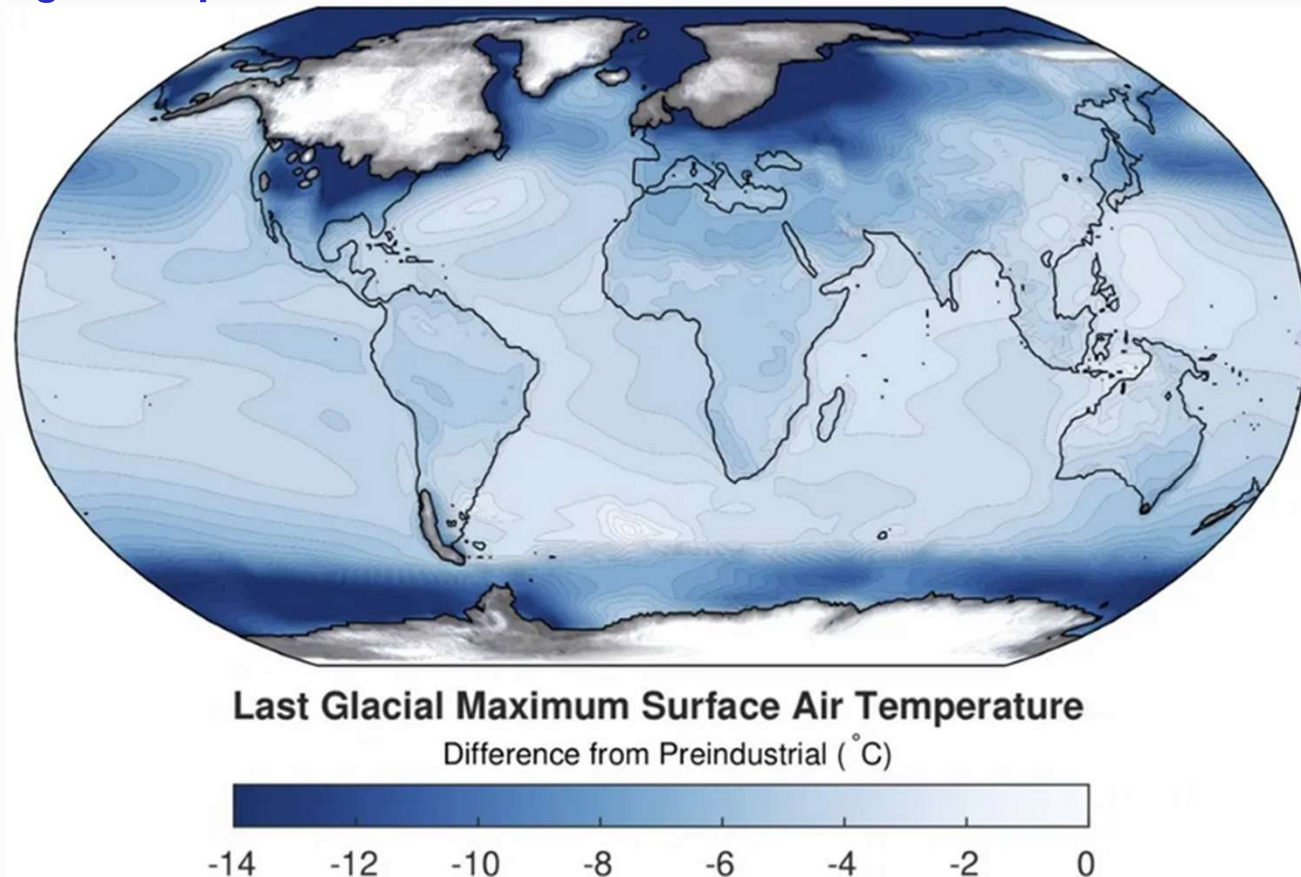
Some large +ve amplifying feedbacks

These feedbacks hasten the transition in and out of Ice ages once the change has been triggered.



Dynamic feedbacks in the climate system

Dynamic feedbacks in the climate system are linked to the speed with which energy is transported from the equator to the pole. During Ice ages polar temperatures were -10°C to -20°C colder than during warm ages. Equatorial temperatures on the other hand were only -2 to -6°C colder than during warm ages indicating the transport of heat from the equator to the pole was suppressed during these periods



This global map indicates the temperature differences between now and preindustrial times, where dark blue translates to cooler temperatures.
Jessica Tierney, via University of Arizona

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 15th September if you want to get them corrected before the Midsem exam

MCQs

1) Assertion (A): The greenhouse effect is primarily due to water vapour in the atmosphere.

Reasoning (R): Water vapour is the most abundant greenhouse gas and traps a significant heat radiated from the Earth's surface.

- a) Both A and R are true, but the R is not the correct explanation of the A.
- b) Both A and R are true, and the R is the correct explanation of the A.
- c) The A is true, but the R is false.
- d) The A is false.

2. The term albedo refers to:

- a) The density of air in the atmosphere
- b) The reflection of solar radiation from Earth's surface
- c) The absorption of carbon dioxide by plants
- d) The amount of ozone in the atmosphere

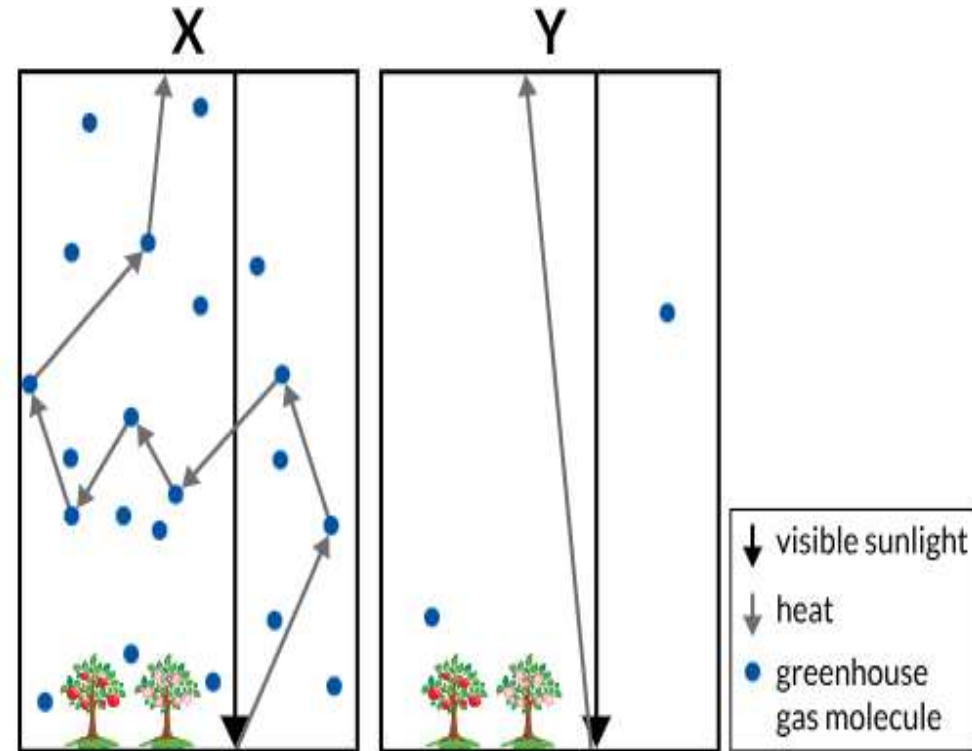
3. In the lower layers of atmosphere, what range of wavelengths of light is predominant?

- a) Less than 100 nm
- b) Greater than 300 nm
- c) Between 100-300 nm
- d) All wavelengths are equally present

MCQs

1) The diagrams on the right show visible light and heat traveling through Earth's atmosphere and interacting with greenhouse gas molecules. Which image best represents an atmosphere in which Earth is experiencing global cooling, and why?

- a) Image X, because a relatively high concentration of greenhouse gases blocks more visible sunlight.
- b) Image X, because a higher concentration of greenhouse gases absorbs more heat than image Y.
- c) Image Y, because a lower concentration of greenhouse gases blocks less visible sunlight.
- d) Image Y, because a lower concentration of greenhouse gases absorbs less heat than Image X.



MCQs

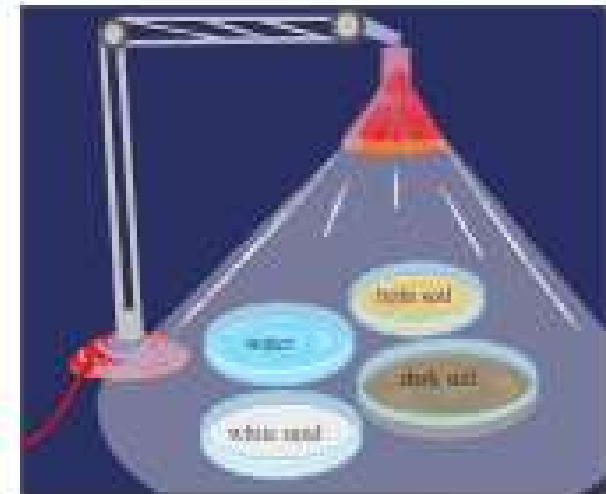
1. Which of the following list of greenhouse gas is most abundant in our atmosphere?
 - a) carbon dioxide CO_2
 - b) methane CH_4
 - c) nitrous oxide N_2O
 - d) SF_6
2. Which of the following correctly describes a positive feedback loop involving melting sea ice?
 - a) surface warming $\rightarrow$ melting sea ice $\rightarrow$ ocean exposure $\rightarrow$ increased sunlight absorption
 - b) surface warming $\rightarrow$ melting sea ice $\rightarrow$ more ocean mixing $\rightarrow$ more severe storms
 - c) surface warming $\rightarrow$ melting sea ice $\rightarrow$ sea level rises $\rightarrow$ increased coastal flooding
 - d) surface warming $\rightarrow$ melting sea ice $\rightarrow$ cooling ocean $\rightarrow$ decreased sunlight absorption
3. Which of the following greenhouse gases is the most abundant in Earth's atmosphere and tends to have the shortest atmospheric lifetime?
 - a) water vapor H_2O
 - b) carbon dioxide CO_2
 - c) methane CH_4
 - d) chlorofluorocarbons (CFCs)
4. Which of the following greenhouse gases has the highest global warming potential and is mostly associated with refrigerants and aerosol chemicals?
 - a) water vapor H_2O
 - b) chlorofluorocarbons (CFCs)
 - c) methane CH_4
 - d) carbon dioxide CO_2

MCQs

1. group of environmental science students designed an experiment to study how different surfaces of the Earth reflect and absorb heat. They set up four identical dishes, each with a different material, and placed them under a lamp (as illustrated below).

Which material will likely have the largest temperature increase during the experiment?

- a) dark soil
- b) light soil
- c) water
- d) white sand



2. Carbon dioxide from the atmosphere mixes with water at the surface of the ocean and is dissolved into the water. There, the dissolved carbon dioxide gas reacts with the liquid water to form carbonic acid. Which of the following statements best describes how carbonic acid impacts the pH of ocean water?

- a) Carbonic acid produces hydrogen ions (H^+) in water, which raises the pH.
- b) Carbonic acid produces hydrogen ions (H^+) in water, which lowers the pH.
- c) Carbonic acid produces hydroxide ions (OH^-) in water, which raises the pH.
- d) Carbonic acid produces hydroxide ions (OH^-) in water, which lowers the pH.

MCQs

- 1) Weather is the _____.
 - a) atmospheric temperature and precipitation averaged over a long period of time
 - b) atmospheric temperature and precipitation during a short period of time
 - c) year to year change in the atmospheric temperature and precipitation .
 - d) Wind and rain
2. _____ can be defined as the weather conditions of a large area for a long period of time
 - a) weather
 - b) Climate
 - c) Seasons
 - d) Climate change
3. When the snowline in the Himalayas shifts upwards over decades this is due to _____?
 - a) Change in weather
 - b) Too less snowfall
 - c) Change in climate
 - d) Change in the atmospheric lapse rate

MCQs

- 1) What is the main difference between weather and climate?
 - a) Climate is always changing weather is not
 - b) weather is always changing climate is not
 - c) Weather is a short-term average, climate is a long-term average
 - d) Weather includes parameters like wind and rain climate does not
2. Which of the following affects a place's climate the most?
 - a) Distance from the equator
 - b) Distance from the ocean
 - c) The type of vegetation that grows
 - d) Daily weather
3. Which term describes a large region with a certain climate and types of plants and animals?
 - a) Biome
 - b) Hadley cell
 - c) Continent
 - d) Storm tracks

Numerical

1. The solar constant for Venus is 2640 W m^{-2} and the planetary albedo of Venus is 0.75. What would its effective temperature be as observed from space? (Stefan Boltzmann constant = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$). Even so, Venus is closer to the Sun than the Earth, in this case why is its effective surface temperature lower than that of the Earth ($T_{\text{eff}} = 255 \text{ K}$)
2. Calculate the effective temperature for snowball earth (Albedo 0.84) 1370 $[\text{W/m}^2] = S_0$

Long answer

1. What criteria determines whether an IR active anthropogenically emitted gas will act as an anthropogenic green house gas in the Earth's atmosphere ? Why does CO which absorbs at $5.26\text{ }\mu\text{m}$ not act as an anthropogenic greenhouse gas?
2. Define greenhouse gas and list the most important natural and anthropogenic greenhouse gases.
3. Define global warming potential! What is the global warming potential of methane for a 20 and 100 year timeframe.
4. Why is the GHG forcing not equally distributed across the globe despite the fact that gasses are well mixed. Why is the forcing in W/m^2 highest in certain tropical areas and lower near the pole. What causes the temperature response to the same forcing to be larger near the poles compared to near the equator?
5. Name the six big positive feedback cycles that make our climate system unstable. Which of these 6 cycles result in larger changes in the local climate of the areas in which the perturbations of the feedback cycle occur. Which of these have primarily global impacts.
6. Explain the difference between weather and climate.